

Question 1 : Object Counting under a Domain Shift

Overview

The image inspection team has received black-and-white images containing squares, triangles, and stars. Your job is to use a CNN model to predict the total number of shapes in each image.

Your training data consists of clean images but all test images have been corrupted in various ways. The base (uncorrupted) images in the test set never appear in the training or validation sets so there is a domain shift that can't be solved by simply memorizing images.

You may use OpenCV to inspect images, perform preprocessing, and conduct data augmentation, but the CNN model must be built with PyTorch.

Description

The source dataset contains 1,795 clean base images. This source dataset is split (and, for the test set, corrupted) into the training set, validation set, and test set. The split is done at the `image_id` level and uses stratification on `total_shapes` to roughly preserve the distribution of object counts, while guaranteeing that `image_id` never repeats across splits.

- `train.csv` and `train_images.npz` : 1,255 clean images, with `total_shapes`
- `validation.csv` and `validation_images.npz` : 270 clean images, with `total_shapes`
- `test.csv` and `test_images.npz` : 1,890 corrupted images from another 270 base images, without `total_shapes`

The CSV files store id and metadata while the image array, ordered by id, sits under the key `images` in the correspondingly named NPZ file.

Public Data

File structure under `data/public` :

```
train.csv
train_images.npz
validation.csv
validation_images.npz
test.csv
test_images.npz
```

`train.csv` and `validation.csv` have columns `id`, `image_id`, `total_shapes`, while `test.csv` has `id` only, with no label.

Evaluation

Evaluated with Mean Absolute Error against the true object counts of the test set.

```
MAE = mean(abs(y_true - y_pred))
```

Submission File

Submit `submission_cv.csv` with 1,890 rows.

```
id,predicted_count
0,24
1,24
2,23
```

- `id` : the image's index in `test_images.npz`
- `predicted_count` : the number of shapes predicted by the model; must be a non-negative integer.



Leagues of Code